

How to Read a Peptide COA: Complete Verification Guide (2026)



A Certificate of Analysis (COA) is your only proof that a peptide contains what the vendor claims. In an unregulated market where **nearly 40% of vendors fail independent purity testing**, knowing how to read and verify a COA isn't optional, it's an essential skill.

This guide will teach you exactly what to look for in a legitimate COA, how to spot fakes, and the steps to verify any certificate before you buy.

What you'll learn:

- The 9 elements every legitimate COA must have
- How to spot fake or suspicious certificates
- Step-by-step verification process
- Quick reference checklist

What is a Certificate of Analysis (COA)

A Certificate of Analysis is a document from a testing laboratory that verifies:



Identity: Is this really BPC-157, or something else?



Purity: What percentage is the target peptide vs. impurities?



Contaminants: Are there harmful substances?



Batch Information: When was it tested? Which production lot?

Think of a COA like a lab report for your peptide. Without one, you're relying on a vendor's word—and when your health is at stake, that isn't enough.

Key Point:

A COA from the vendor's own lab is better than nothing, but third-party testing from an independent lab is the absolute gold-standard

The 9 Elements of a Legitimate COA

Not all COAs are created equal. Here's what a trustworthy certificate must include:

1. Testing Laboratory Information

What to look for:

- Lab name and full address
- Contact information (phone, email, website)
- Lab director or analyst signature
- Accreditation status (ISO 17025 is ideal)



RED FLAG: No lab name, generic headers like "Quality Control Department," or untraceable contact information.

2. Product Identification

What to look for:

- Peptide name (e.g., "BPC-157 Acetate")
- Chemical formula or sequence
- CAS number (if applicable)
- Molecular weight



RED FLAG: Vague descriptions like "Research Peptide" without specific identification.

3. Batch/Lot Number

What to look for:

- Unique identifier that matches your vial exactly
- Format varies by lab (e.g., "LOT-2026-0115-BPC")



RED FLAG: Generic numbers like "SAMPLE001," "N/A," or the same lot number appearing on COAs for different products.

4. Date of Analysis

What to look for:

- Testing date within the past 12 months
- Ideally within 90 days of your purchase



RED FLAG: Dates from 2024 or earlier, no date listed, or future dates (yes, this happens).

5. HPLC Purity Results

What to look for:

- Purity percentage (98%+ is the standard for quality peptides)
- Testing method specified (e.g., "RP-HPLC, C18 column")
- Retention time noted



RED FLAG: Purity listed without methodology, or suspiciously perfect numbers like "100.00%."

6. Mass Spectrometry (MS) Confirmation

What to look for:

- Observed molecular weight matching theoretical weight
- MS method specified (ESI-MS, MALDI-TOF, etc.)
- Mass spectrum image or data



RED FLAG: No mass spectrometry data at all. HPLC alone cannot confirm identity —only purity.

7. Chromatogram

What to look for:

- Visual graph showing separation peaks
- Dominant single peak for target peptide
- Minimal secondary peaks (impurities)
- Labeled axes and retention time



RED FLAG: No chromatogram provided, blurry/low-resolution images, or chromatograms that look copy-pasted.

8. Additional Testing (When Applicable)

For injectable peptides, look for:

Test	Purpose	Standard
Endotoxin (LAL)	Bacterial contamination	< 0.5 EU/mg
Sterility	Microbial contamination	Pass/Fail
Heavy Metals	Lead, arsenic, mercury	Below detection limits



RED FLAG: Injectable products with no sterility or endotoxin testing mentioned.

9. Pass/Fail Determination

What to look for:

- Clear specification limits
- Actual results vs. acceptable range
- Overall pass/fail statement



RED FLAG: Results without reference ranges, or missing conclusion.

HPLC vs. Mass Spectrometry: Why You Need Both

This is the most misunderstood aspect of peptide testing—and one of the easiest ways to spot an incomplete COA.

What HPLC Tells You (And What It Doesn't)

High-Performance Liquid Chromatography (HPLC) separates compounds in a sample based on how they interact with a solvent. The result is a chromatogram—a graph showing peaks for each compound detected.

What HPLC measures:

- The percentage of the main compound vs. impurities
- How “clean” the sample is overall
- Relative amounts of different substances present

What HPLC cannot do:

- Confirm the identity of the compound
- Tell you *what* the main peak actually is
- Distinguish between structurally similar peptides

Think of HPLC like weighing a bag of coins. It can tell you the bag is 99% full of one type of coin—but it can't tell you whether those coins are quarters or nickels.

What Mass Spectrometry Tells You

Mass Spectrometry (MS) measures the molecular weight of compounds in a sample. Each peptide has a unique molecular weight based on its amino acid sequence.

What MS measures:

- The exact molecular weight of the compound
- Whether the peptide matches its expected formula
- Detection of truncated or modified sequences

What MS cannot do:

- Measure purity or concentration
- Tell you how much of the compound is present
- Detect impurities that don't ionize well

Mass spectrometry is like checking the serial number on those coins. It confirms you actually have quarters—but doesn't tell you if any are counterfeit or how many you have.

Why You Need Both: A Real-World Example

Method	What it measures	Limitation
HPLC	Purity (% of target compound)	Cannot confirm identity
MS	Identity (molecular weight)	Doesn't measure purity

Scenario: A vendor sells “BPC-157” that’s actually a truncated fragment missing the last two amino acids.

Test	Result	What it means
HPLC	✓ 99.2%	The sample is very pure, but pure what?
MS	✓ 1419 Da (expected 1419 Da)	Correct! This is actually BPC-157

Now consider a **bad scenario**:

Test	Result	What it means
HPLC	✓ 99.2%	Looks great on paper
MS	✗ 1206 Da (expected 1419 Da)	Wrong molecule—missing ~2 amino acids

The HPLC passed because the sample was pure. But pure doesn't mean correct. Without mass spectrometry, you'd never know you received the wrong peptide.

Common Shortcuts to Watch For

Some vendors cut costs by only providing HPLC data:

- **“98% purity” with no MS data** — You have no proof of identity
- **MS data from a different batch** — Purity may vary between lots
- **In-house HPLC only** — No independent verification of either metric

Bottom Line: A COA with only HPLC data is like a report card with only attendance—it tells you something, but not the full picture. Always look for both HPLC (purity) and Mass Spec (identity) on the same certificate, from the same batch.

Purity Standards: What’s Acceptable?

Purity Level	Quality Grade	Recommendation
≥99%	Premium/Pharmaceutical	Ideal for all applications
98-99%	Research grade	Acceptable standard
95-98%	Lower research grade	Proceed with caution
< 95%	Substandard	Avoid

Most reputable vendors consistently deliver 98%+ purity. If a vendor’s COAs regularly show 95% or below, consider alternatives.

How to Verify a COA Is Real

A COA is only valuable if it's authentic. Here's how to verify:

Step 1: Check the Testing Lab

Search for the lab name online:

- Does their website exist?
- Do they actually test peptides?
- Can you find their address and contact information?

Trusted third-party labs include:

- [Janoshik Analytical](#) (Slovakia) – Offers online verification
- Colmaric Analyticals (USA)
- [MZ Biolabs](#)
- Freedom Analytics Labs (USA)

Step 2: Match the Batch Number

The lot number on the COA must exactly match the lot number on your vial. If they don't match, the COA doesn't apply to your product.

Step 3: Confirm the Report with the Lab

Online Verification (fastest when available)

- Janoshik: Enter the report ID at janoshik.com/verify to confirm authenticity
- Some labs have QR codes or verification portals on their COAs

Email Verification if no online portal exists. Email the lab with:

- The Report ID or document number
- Date of analysis
- Product tested
- Vendor name

Step 4: Request the COA Before Ordering

The best time to verify a COA is before you spend any money. Quality vendors have nothing to hide—they'll happily provide batch-specific certificates for any product you're considering. Simply email or use their live chat to request the COA for the exact lot number currently in stock.

If a vendor hesitates, only offers COAs after purchase, or sends you a generic "sample" certificate that isn't tied to a real batch, take your business elsewhere. Transparency about testing is the bare minimum for any reputable supplier, and vendors who make verification difficult are often the ones with something to hide.

Red Flags: How to Spot a Fake COA

Watch for these warning signs:

Document Red Flags

- Missing batch/lot numbers
- Outdated testing (more than 12 months old)
- No chromatogram or mass spec data
- Blurry, low-resolution scans
- Purity listed as exactly 100% (unrealistic)

Behavioral Red Flags

- Same COA used for multiple batches
- COA provided only after purchase and complaint
- Vendor refuses to share testing data
- Lab cannot verify the document when contacted

The “Template COA” Problem

Some vendors create template COAs with impressive-looking data, then reuse them across all batches. To spot these:

- Request COAs for two different batch numbers
- Compare the documents—are they identical except for the lot number?
- Check if the testing dates make sense for when the batches were produced

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